Poroelastic rupture: equations, weak form, SIPG and code map

Biot's equations in the Nitsche DG solver, from the strong form to the code · revision 1.2, 20 Oct 2026

Revision 1.2. Answers Chunhui Zhao's three questions of 20 Oct 2026 and records the decision on F1. New §4.6 compares the strong form with the three-field (u, w, p) form of the note Chunhui Zhao shared (Sivaselvan et al. 2016; Mella et al. 2023): it is the same physics, and the note has two typos. New §9.5 says how SeisSol steps the drag: it does not use γ_w, which is our own. New §11 says where every equation comes from, paper by paper, and marks the ones derived here. Biot (1956, 1962) is now cited for the strong form (§4.2). F1 is decided (§1.3); F2 and F3 still need Chunhui Zhao's decision. Companion to plan revision 3 (`poro-plan-rev3.pdf`), which holds the implementation steps and the tests. The checks named in the text now point to the unit tests of plan revision 3 (U2-5, U2-13, U2-16, U2-18) and its quasi-static tests V-Q1–V-Q3. Revisions 1 and 1.1 are kept.

Revision 1.1. Adds a general fluid source, the injection rate $Q(x,t)$, to the fluid mass balance (Chunhui Zhao, 20 Oct 2026). SeisSol's `tpv5-poro` starting pressure becomes a special case of the same equations, used for the comparison (§4.2, §6.5, §7.4). Revision 1 is kept. Companion to plan revision 2.2 (`poro-plan-rev2.2.pdf`). It derives every equation the poroelastic solver needs, from the strong form through the element weak form and SIPG to the matrices and the time step, and names the file and function that will implement each one. Nothing is implemented. Chunhui Zhao reviews this document before implementation is approved. Code citations are at `5b2c77d` (branch `nitsche-dynamic-rupture-poroelastic`; its code is identical to trunk `a4c9dee`). Writing the derivation out in full turned up three changes to the plan's formulas. They are F1–F3 in §1.3, and each needs Chunhui Zhao's decision.

1. What this document answers

The poroelastic solver adds one unknown field, the relative fluid displacement w, in the same DG space as the solid displacement u. Everything the elastic solver has stays in place: the SIPG elastic form, the fault traction kernel, the slip load, the friction closure and the explicit step. The new physics enters in four places:

1. **A mass and drag that couple u and w inside each element.** They are element-local, so the step stays explicit (§8.2).
2. **New SIPG blocks built from divergences.** The solid block is today's elastic SIPG form with the undrained modulus $\lambda_u = \lambda + \alpha^2 M$. One new form, used three times, couples $\nabla\cdot u$ with $\nabla\cdot w$ and penalizes only the normal jump of w (§6).
3. **A fault pressure taken from the fluid block's numerical flux.** It lowers the effective normal stress the friction law sees (§7).
4. **Fluid sources.** An injection rate $Q(x,t)$ adds fluid and loads both the fluid and the solid rows. SeisSol's starting pressure for `tpv5-poro` is a special case that loads only the fluid rows (§6.5).

Section 1.1 gathers the equations, §1.2 says where each one lives in the code, and §2 is the code map. The derivations follow in §3–§10 and are cited by equation number.

1.1 The equations at a glance

Strong form (§4). In Ω, with the unknowns u (solid displacement) and $w = \phi(U_f - u)$ (relative fluid displacement), measured from the state at $t = 0$:

$$\begin{aligned} \rho\ddot{u} + \rho_f\ddot{w} - \nabla\cdot\sigma &= f, & \sigma &= C_d : \varepsilon(u) - \alpha\,(p - p_{\text{init}})\,I,\\ \rho_f\ddot{u} + m\ddot{w} + b\dot{w} + \nabla p &= 0, & p &= p_{\text{init}} + M\big(\zeta_f - \alpha\nabla\cdot u\big),\\ \partial_t\zeta_f + \nabla\cdot\dot{w} &= Q, & \zeta &= \alpha\nabla\cdot u + \nabla\cdot w, \quad V = \textstyle\int_0^t Q\,dt. \end{aligned}$$

Integrated in time, $p = p_{\text{init}} + M(V - \zeta)$ and $\sigma = C_d : \varepsilon(u) + \alpha M(\zeta - V)I$. Q is the injected fluid volume per unit rock volume per second; p_{init} is a starting pressure with no strain and no stress, zero in general and SeisSol's Gaussian for `tpv5-poro`. On the permeable fault: $[\![u]\!]\cdot n = 0$, $[\![u]\!] = d$ (slip, tangential), $[\![\sigma n]\!] = 0$, $[\![p]\!] = 0$ and $[\![w\cdot n]\!] = 0$. Friction sees $\sigma_n' = -(T + S)\cdot n - (p_0 + p)$, compression positive, with $T = \sigma n$.

Semi-discrete SIPG form (§6–§8). Find $x = [u; w]$ in the vector DG space such that

$$\mathbb{M}\ddot{x} + \mathbb{D}\dot{x} + \begin{pmatrix} K_E^u & Z_{uw} \\ Z_{wu} & Z_{ww} \end{pmatrix} x = \begin{pmatrix} F_E(t) + G^u d + F_{V,u}(t) \\ F_{V,w}(t) + F_{\text{init}} \end{pmatrix},$$

where $\mathbb{M} = M_\rho \otimes M_e \otimes I_3$ and $\mathbb{D} = D_\rho \otimes M_e \otimes I_3 + C$ are element-local, K_E^u and G^u are today's elastic SIPG stiffness and slip load built with (λ_u, μ), the Z blocks come from one new divergence form (6.3), and the source loads F_V, F_{init} from (6.6).

Time step (§9). Newmark with $\beta = 0$ and a velocity-coefficient matrix Γ: $\frac{1}{2}$ for u and the fitted $\gamma_w(z) = 1/(1 - e^{-z}) - 1/z$ for w, with $z = \Delta t/\tau$:

$$x^{n+1} = x^n + \Delta t\, v^n + \tfrac{\Delta t^2}{2} a^n, \qquad (\mathbb{M} + \Delta t\, \mathbb{D}\Gamma)\, a^{n+1} = F^{n+1} - \mathbb{K} x^{n+1} - \mathbb{D} v^*, \qquad v^{n+1} = v^* + \Delta t\, \Gamma a^{n+1}.$$

1.2 Where each equation lives

New files are under `miniapps/seas/multiphysics/poroelastic/` (plan §4.1). Existing files are at `5b2c77d`, under `miniapps/seas/`.

Eq.	What it is	New function (file)	Existing code it builds on
(4.4)	Derived constants $\rho, m, \alpha, M, \lambda_u, b, \tau$	`PoroMaterialSample` (`poro_material.hpp`)	`NitscheMaterialSample` (`domain/nitsche_material_sample.hpp`)
(6.3)	Solid block a_E^u: today's SIPG with λ_u	none: build the operator from an undrained sample (λ_u, μ, ρ)	`ElasticityDomainOperator`, `DGElasticityIPCombinedIntegrator` (`integrator/dg_elasticity_ip_combined_integrator.hpp:58-200`)
(6.3)	Divergence form B_c: coupling blocks ($c = \alpha M$) and fluid block ($c = M$)	`DGPoroDivIntegrator` (`dg_poro_div_integrator.hpp`)	Same (c_0, c_1, c_2) pattern as `AssembleFaceBlock` (`…combined_integrator.hpp:889-947`)
(6.4)	Solid penalty σ_u	none (reused with λ_u)	`ComputePenalty` (`…combined_integrator.hpp:857-881`)
(6.5)	Fluid penalty σ_w	`DGPoroDivIntegrator::Penalty`	The same formula with $(D\lambda + 2\mu)^2/(2\mu)$ replaced by M
(6.6)	Source loads: injected volume $V(t)$ on both rows, starting pressure on the fluid rows	`DGPoroDivIntegrator::AssembleSourceRHS`; `PoroSource` (`poro_source.hpp`) for Q and V	Like the Dirichlet load, assembled every step

Eq.	Content	Source	Ours?
(7.1)	Slip load $G^u d$	none (reused with λ_u)	`AssembleSlipFaceRHS` (`…combined_integrator.hpp:212-330`); `SlipLoad` (`domain/nitsche_wave_operator.hpp:470`)
(7.2)	Fault pressure $\hat{p}$	`FacePressure` (`poro_fault_pressure.hpp`)	Quadrature-point loop of `ComputeTractionAtQuadPoints` (`…combined_integrator.hpp:435-556`)
(7.3)	Total traction: today's kernel plus $\{\{\alpha M(\nabla \cdot w - V)\}\} n$	`CouplingTraction` (`poro_fault_pressure.hpp`)	`ComputeTractionDiagnostics`, box 6
(7.4)	Effective normal stress	hook H2: `FaultPressureHook`	`AdvanceFault_` boxes 6–9, `CloseDOF` (`domain/nitsche_wave_operator.hpp:993-1099, 488-516`)
(8.3)	Element mass and drag inverse	`PoroMass::Apply` (`poro_mass.hpp`)	`NitscheMass` (`domain/nitsche_mass.hpp:50-205`)
(9.1)–(9.2)	The time step	`PoroNewmarkStepper::Step` (`poro_newmark.hpp`)	`ImplicitSolve`, `Acceleration_` (`domain/nitsche_wave_operator.hpp:278-316, 840-909`)
(9.4)	Fitted coefficient γ_w	`FittedGamma(z)` (`poro_newmark.hpp`)	none
(9.6)	Time-step bound	`PoroMaxDt` (`poro_cfl.hpp`)	`ComputeMaxDt` (`solver/nitsche_cfl.hpp:68-92`)
(10.2)–(10.3)	Quasi-static Biot and the split step (P7)	`PoroQDOperator`, `PoroDiffusionSolver`	`SEASQuasiDynamicOperator::Mult` (`solver/seas_operator.hpp:318`)

1.3 Three changes to the plan's formulas

All three come from writing the derivation out in full. F1 is decided (Chunhui Zhao, 20 Oct 2026). F2 and F3 need Chunhui Zhao's decision before implementation.

F1. Fluid sources: the injection rate in general, SeisSol's start as a special case (a correction). In general, fluid enters through an injection rate $Q(x,t)$ in the fluid mass balance (Chunhui Zhao, 20 Oct 2026). Integrated in time, it raises the pressure by MV, with $V = \int_0^t Q\, dt$, and it changes the total stress by $-\alpha M V I$, so it loads the solid too (§4.2). SeisSol's `PressureInjection` is not a source. It starts from zero strain and zero total stress with a pore pressure p_{init} (SeisSol docs, `docs_initial-condition.rst:146-150; parameters.par` has no source). That is the special case $Q = 0$

with a starting pressure, which loads only the fluid. Plan revision 2.2 (§8.2) instead put the pressure inside $\sigma = C_d : \varepsilon - \alpha p I$, which is the instant injection $Q = (p_{\mathrm{inj}}/M)\,\delta(t)$, not SeisSol's start. On the fault an instant injection at first lowers the effective normal stress by only $(1-\alpha)p_{\mathrm{inj}}$. In `tpv5-poro` ($\alpha = 0.523$, 45.2 MPa at the hypocentre) it would give $\sigma_n' = 98.4$ MPa instead of SeisSol's 74.8 MPa at $t = 0$, and an overstressed patch of radius 1.6 km instead of 3.2 km. So the comparison with SeisSol uses the special case (§4.2, §6.5, §7.4). **Required** to match SeisSol. **Decided** by Chunhui Zhao on 20 Oct 2026: "in general, injection rate is more general", with the `tpv5-poro` start as the special case for the comparison.

F2. Keep the total traction together in the SIPG fluxes. Plan revision 2.2 (eq. 8.3b) splits off the pressure part of the stress and penalizes $[\![(\alpha u + w)\cdot n]\!]$. Where α jumps between elements, the exact solution makes that jump nonzero, so the face terms are inconsistent by an amount that does not shrink with the mesh (Proposition 6.1). Keeping the total traction σn as the solid flux and the pressure p as the fluid flux is consistent for any jumps in α, M, λ, μ. It also makes the solid block exactly today's elastic SIPG form with (λ_u, μ): stiffness, penalty, traction kernel and slip load are reused unchanged, and with the fluid locked ($w = 0$) the operator is today's elastic operator bit for bit (Proposition 6.2). That is what the plan's undrained-equivalent TPV205 test needs: today the plan matches only the Kelvin–Voigt layer to the elastic gold (plan revision 2.2, §8.2 remark); with F2 the whole solid operator matches. The two benchmarks have uniform α, so for them F2 changes the penalties but not the consistency. **Recommended.**

F3. Let the solid velocity see the fitted fluid update. With the plan's diagonal $\Gamma = \mathrm{diag}(\frac{1}{2}, \gamma_w)$, the discrete total momentum $\rho\dot{u} + \rho_f \dot{w}$ gains $\rho_f \Delta t(\gamma_w - \frac{1}{2})(a_w^{n+1} - a_w^n)$ per step. A non-diagonal Γ, whose u row is $[\frac{1}{2},\ \frac{\rho_f}{\rho}(\frac{1}{2} - \gamma_w)]$, advances total momentum by the trapezoid rule and leaves the fitted fluid update exact. It costs nothing, because the matrix to invert is unchanged away from absorbing boundaries (§9.3). The stepper stays the fitted Newmark stepper of decision D2. **Recommended.** Tests U2-16 (energy) and U2-18 (stability) of plan revision 3 check its energy behaviour and stability.

2. Code map

2.1 Files

```
miniapps/seas/multiphysics/poroelastic/
  poro_config.hpp/.cpp         parse [multiphysics.poroelastic]; every key required
  poro_material.hpp            PoroMaterialSample: eq. (4.4) per element, and the
                               undrained sample (lambda_u, mu, rho) for the solid block
  poro_dispersion.hpp          plane-wave speeds, eq. (4.6) (CFL and tests)
  dg_poro_div_integrator.hpp   B_c of (6.3), penalty (6.5), source loads (6.6)
  poro_source.hpp              PoroSource: Q(x, t), V = int Q dt, p_init; (4.3), (6.6)
  poro_mass.hpp                (M + dt D Gamma)^{-1}, eq. (8.3)
  poro_nitsche_operator.hpp    owns NitscheWaveOperator (undrained); force of (9.2)
  poro_newmark.hpp             PoroNewmarkStepper, FittedGamma, eqs. (9.1)-(9.5)
  poro_fault_pressure.hpp      FacePressure (7.2), CouplingTraction (7.3),
                               FaultPressureHook (7.4)
  poro_cfl.hpp                 PoroMaxDt, eq. (9.6)
  poro_receivers.hpp           off-fault receivers: v, w_dot, p, sigma
  poro_qd_operator.hpp         P7: split step, eqs. (10.2)-(10.3)
  poro_diffusion_solver.hpp    P7: the SPD solve (10.3), cached MUMPS + PCG
  poro_handoff.hpp             P8: state copy, §10.3
```

Plan revision 2.2 §4.1 named the integrator `dg_poro_zeta_integrator.hpp`; with F2 it carries the divergence form B_c, hence the new name.

2.2 One dynamic step, with equation numbers

New work is in **orange**. Boxes 3–11 are today's fault advance (`domain/nitsche_wave_operator.hpp:28-43, 993-1099`), run by the elastic operator built from the undrained sample.

```
PoroNewmarkStepper::Step(x = [u; w], v, a, t, dt)
  1  x^{n+1} = x^n + dt v^n + dt^2/2 a^n                          (9.1)
  2  v* = v^n + dt (I - Gamma) a^n                                 (9.1), Gamma from (9.5)
  2b V^{n+1} = V^n + dt/2 (Q^n + Q^{n+1}), or exact                (6.6)
  3  op.AdvanceFault(x^{n+1}, t_{n+1}, v*)                        boxes 3-11, changed only in:
       6  T_h = ComputeTractionDiagnostics(u^{n+1}, d*)           existing kernel, lambda_u
       6b T = T_h + avg(alpha M (div w^{n+1} - V^{n+1})) n        (7.3)
       6c p_hat = FacePressure(u^{n+1}, w^{n+1})                   (7.2)
       8  sigma_n' = -(T + S).n - (p_0 + p_hat)                    (7.4)
       9  closure, gamma = sigma_u dt/2 (unchanged code)           Proposition 7.1
  4  F_u = F_E(t_{n+1}) + G d^{n+1} - K_E u^{n+1}                 existing Acceleration_ force
     F_u += F_V,u(t_{n+1}) - Z_uw w^{n+1}                          (6.3), (6.6)
     F_w  = F_V,w(t_{n+1}) + F_init - Z_wu u^{n+1} - Z_ww w^{n+1}  (6.3), (6.6)
     F -= D v*  (drag on w; dashpot C on u as now)                 (9.2)
  5  a^{n+1} = PoroMass(Gamma).Apply(F)                            (8.3), (9.2)
  6  v^{n+1} = v* + dt Gamma a^{n+1}                               (9.1)
```

2.3 The new functions

Function	What it does, in plain words	Inputs → outputs	Implements
`PoroMaterialSample`	Reads the ten Biot inputs at each element centroid and derives everything else once. It also hands the elastic operator an undrained sample, with λ_u in place of λ and the bulk density ρ.	mesh, material map → per element $\rho, \rho_f, m, b, \alpha, M, \lambda_u, \tau, c_{\text{fast}}$	(4.4)
`DGPoroDivIntegrator::AssembleElementMatrix`	The volume part of B_c: coefficient times divergence of the test times divergence of the trial.	element, c → $3n_e \times 3n_e$ block	(6.3), volume
`DGPoroDivIntegrator::AssembleFaceMatrix`	The face part of B_c: consistency and symmetry on normal jumps, plus the penalty for the fluid block. The same (c_0, c_1, c_2) pattern per side pair as the elastic face block.	face, both sides' c (and M) → face block	(6.3) faces, (6.5)
`DGPoroDivIntegrator::AssembleSourceRHS`	The source loads: the injected volume V on both rows, the starting pressure on the fluid rows.	$V(t)$, p_{init} at quadrature points → $F_V(t)$, F_{init}	(6.6)
`PoroSource`	Evaluates the injection rate $Q(x,t)$ from the deck and keeps $V = \int_0^t Q\,dt$ at the quadrature points: trapezoid rule in time, or exact when Q has a closed-form integral. A well is a point source $Q_w(t)\,\delta(x - x_w)$.	deck, t → Q, V, p_{init}	(4.3), (6.6)
`PoroMass::Apply`	Applies the inverse of the element mass plus drag: one 2×2 inverse per element times the existing M_e^{-1}.	force F → acceleration a	(8.3)
`FacePressure`	Evaluates the fault pressure from the fluid block's own flux at the fault quadrature points and projects it to fault DOFs.	$[u; w]^{n+1}$ → $\hat{p}$ per fault DOF	(7.2)
`CouplingTraction`	Adds the fluid part of the total traction, $\{\{\alpha M(\nabla \cdot w - V)\}\}n$, to today's traction kernel output.	w^{n+1}, V^{n+1} → traction increment per fault DOF	(7.3)
`PoroNewmarkStepper::Step`	The explicit step of §2.2.	$x, v, a, t, \Delta t$ → advanced state	(9.1)–(9.5)
`FittedGamma(z)`	γ_w, with a series for small z to avoid cancellation.	z → γ_w	(9.4)
`PoroMaxDt`	Today's CFL formula with the fast P speed.	sample, mesh, p → Δt	(9.6)

Pseudocode of the face block of $B_c(a, b)$, with trial field a and test field b (each u or w), side signs $s_+ = 1$, $s_- = -1$, test on side i and trial on side j, for interior and permeable fault faces:

```
for each quadrature point (weight wq, unit normal n, face Jacobian |J_F|):
  sw = Penalty(M_+, M_-, ...)    fluid block (a = w, b = q) only          (6.5)
  for each side pair (i, j):
    Ntest_i  = s_i  b_i.n            normal jump of the test, side i part
    Ntrial_j = s_j  a_j.n            normal jump of the trial, side j part
    Dtest_i  = c_i  div b_i          c = alpha M (coupling) or M (fluid)
    Dtrial_j = c_j  div a_j
    block(i,j) += wq |J_F| [ -1/2 Ntest_i Dtrial_j      consistency (c0 = -s_i/2)
                             -1/2 Dtest_i Ntrial_j      symmetry    (c1 = -s_j/2)
                             + sw Ntest_i Ntrial_j ]    penalty     (c2 = s_i s_j sw)
```

3. Notation, conventions and assumptions

- **Coordinates.** Global X, Y, Z. On the fault, the local frame is (normal, dip, strike) from `FaultBasis`, with tangent 1 = dip and tangent 2 = strike, as everywhere in the project. Nothing below assumes a planar or vertical fault.
- **Signs.** Stress is tension positive. Pore pressure p is positive in compression. The effective normal stress σ'_n the friction law sees is compression positive, as in the code (`domain/nitsche_wave_operator.hpp:37`).
- **Faces.** Each face F has a $+$ side (MFEM's element 1) and a $-$ side, and n is the unit normal out of the $+$ side. $[\![a]\!] = a^+ - a^-$ and $\{\{a\}\} = \frac{1}{2}(a^+ + a^-)$, as in the code (`…combined_integrator.hpp:514-552`). The scalar normal jump $[\![a \cdot n]\!] = [\![a]\!] \cdot n = a^+ \cdot n^+ + a^- \cdot n^-$ does not depend on which side is $+$.
- **Moduli.** C_d is the drained (frame) stiffness with Lamé moduli λ, μ; C_u is the undrained stiffness with $\lambda_u = \lambda + \alpha^2 M$ and the same μ, so $C_u : \varepsilon(u) = C_d : \varepsilon(u) + \alpha^2 M (\nabla \cdot u) I$.
- **Mesh and space.** Tetrahedra only. V_h is the vector L2 DG space of degree p already used for u; w lives in the same space.

Assumptions, cited by label below:

- **(A1)** Small strain; isotropic drained frame.
- **(A2)** Biot's low-frequency theory: Darcy drag $b\dot{w}$ with $b = \eta/\kappa$, valid below the Biot frequency f_c (51 kHz in `tpv5-poro`, plan §3.3).
- **(A3)** Material constants are constant on each element; they may jump between elements.
- **(A4)** The background state (total prestress S on the fault, pore pressure p_0) is in equilibrium on its own and drives no flow. The unknowns are changes from it, so S and p_0 appear only in the friction law.

> **Lemma 3.1 (face sums).** For piecewise smooth p (scalar) and a (vector), with n_K the outward normal of element K,
>
> $$\sum_K \int_{\partial K} p\, a \cdot n_K = \sum_{F \text{ interior}} \int_F \Big(\{\{p\}\} [\![a \cdot n]\!] + [\![p]\!] \{\{a\}\} \cdot n \Big) + \sum_{F \subset \partial\Omega} \int_F p\, a \cdot n. \tag{3.1}$$
>
> The same holds with $p\,a$ replaced by $(\sigma n_K) \cdot v$: the face term is $\{\{\sigma n\}\} \cdot [\![v]\!] + [\![\sigma n]\!] \cdot \{\{v\}\}$.

Proof. An interior face is seen from both sides, with $n_{K^+} = n$ and $n_{K^-} = -n$, so it contributes $p^+ a^+ \cdot n - p^- a^- \cdot n = [\![pa]\!] \cdot n$. Expanding both sides shows $[\![pa]\!] = \{\{p\}\} [\![a]\!] + [\![p]\!] \{\{a\}\}$. ∎

4. Strong form

4.1 Unknowns and material

The unknowns are the solid displacement u and the relative fluid displacement $w = \phi(U_f - u)$, where U_f is the fluid displacement. The fluid flux relative to the rock is $\dot{w}$ (SeisSol's relative fluid velocity). The ten inputs are those of SeisSol, under the same names, and the derived constants use the same formulas as SeisSol's `Helper.h` (master, lines 41–65, read 20 Oct 2026). The last column is `tpv5-poro` (`material.yaml`).

Input (SeisSol name)	Symbol	Derived constant	Formula	`tpv5-poro`
`bulk_solid`	K_s	bulk density	$\rho = (1-\phi)\rho_s + \phi\rho_f$	1664 kg/m³
`rho` (solid grains)	ρ_s	apparent fluid mass	$m = T\rho_f/\phi$	5200 kg/m³
`lambda`, `mu` (drained frame)	λ, μ	Biot coefficient	$\alpha = 1 - K_d/K_s$, $K_d = \lambda + \frac{2}{3}\mu$	0.523
`porosity`	ϕ	Biot modulus	$M = K_s/\left(1 - \phi - K_d/K_s + \phi K_s/K_f\right)$	6.02 GPa
`permeability`	κ	drag	$b = \eta/\kappa$	1.67 GPa·s/m²
`tortuosity`	T	undrained Lamé modulus	$\lambda_u = \lambda + \alpha^2 M$	6.92 GPa
`bulk_fluid`	K_f	Skempton coefficient	$B = \alpha M/(\lambda_u + \frac{2}{3}\mu)$	0.28
`rho_fluid`, `viscosity`	ρ_f, η	relaxation time	$\tau = (\rho m - \rho_f^2)/(b\rho)$, eq. (8.4)	2.73 µs

The formula for M is the same as $1/M = \phi/K_f + (\alpha - \phi)/K_s$. `PoroMaterialSample` evaluates these once per element (A3); together they are eq. (4.4) below.

4.2 Balance laws and constitutive relations

Total momentum, the relative fluid motion with Darcy drag (A2), the fluid mass balance and Biot's constitutive law are

$$\rho\ddot{u} + \rho_f\ddot{w} - \nabla\cdot\sigma = f, \tag{4.1}$$

$$\rho_f\ddot{u} + m\ddot{w} + b\dot{w} + \nabla p = 0, \tag{4.2}$$

$$\begin{aligned} &\partial_t\zeta_f + \nabla\cdot\dot{w} = Q, && \text{fluid mass balance}\\ &\sigma = C_d : \varepsilon(u) - \alpha\,(p - p_{\text{init}})\,I, \qquad p = p_{\text{init}} + M\big(\zeta_f - \alpha\nabla\cdot u\big), && \text{Biot}\\ &p = p_{\text{init}} + M\big(V - \zeta\big), \qquad \sigma = C_d : \varepsilon(u) + \alpha M\big(\zeta - V\big)I, && \text{integrated, with}\\ &V(x,t) = \int_0^t Q\,dt, \qquad \zeta(u,w) = \alpha\nabla\cdot u + \nabla\cdot w. \end{aligned} \tag{4.3}$$

$$\rho,\ m,\ b,\ \alpha,\ M,\ \lambda_u \text{ from the table in §4.1, per element.} \tag{4.4}$$

> **Where (4.1)–(4.3) come from.** They are Biot's equations in the low-frequency range, with the solid displacement and the relative fluid displacement as unknowns (Biot 1956, 1962). We write them as Antonietti, Botti and Mazzieri (2022) do in their review, eq. (31), checked on 20 Oct 2026: their β is our α, their m (Biot modulus) is our M, their $\rho_w = a\rho_f/\phi$ is our m, and their η/k is our b. The injection rate Q and its time integral V are the general fluid source Chunhui Zhao asked for (F1). §11 lists the source of every equation.
>
> Biot (1956), J. Acoust. Soc. Am. 28(2) 168–178; Biot (1962), J. Appl. Phys. 33(4) 1482–1498; Antonietti, Botti, Mazzieri (2022), Vietnam J. Math. 50(4) 997–1028, §5.

σ is the total stress change and p the pore pressure, both measured from the background (A4). ζ_f is the increment of fluid content, the fluid volume gained per unit rock volume. Q is the injected fluid volume per unit rock volume per second (negative for extraction); a well is the point source $Q = Q_w(t)\,\delta(x - x_w)$. Integrating the mass balance from $t = 0$, when $u = w = 0$ and $\zeta_f = 0$, gives $\zeta_f = V - \nabla\cdot w$, which is the integrated line. So ζ measures how much the pores are squeezed: rock compression ($\nabla\cdot u < 0$) or fluid leaving ($\nabla\cdot w > 0$) both raise p, and injected fluid raises it by MV.

> **Two ways to put fluid in (finding F1).** Biot's law fixes rates, $\dot\sigma = C_d : \dot\varepsilon - \alpha\dot p I$ and $\dot p = M(\dot\zeta_f - \alpha\nabla\cdot\dot u)$, so the state at $t = 0$ enters only as integration constants. In general that state is the background, so $p_{\text{init}} = 0$, and fluid enters through Q. Injected fluid raises p by MV and, through $-\alpha p$, changes the total stress by $-\alpha MVI$: the skeleton feels the fluid pushed into it. SeisSol's `PressureInjection` is different. It sets $p(0) = p_{\text{init}}(x) = m_0\exp(-w_0|x - x_0|^2)$ and every other quantity to zero, so $\sigma(0) = 0$, and `parameters.par` has no source. In `tpv5-poro`, $m_0 = 50$ MPa, $w_0 = 10^{-7}$ m^{-2} and x_0 is 1 km off the fault at 7.5 km depth. For the comparison with SeisSol we therefore take $Q = 0$ and this p_{init}. Plan revision 2.2 wrote $\sigma = C_d : \varepsilon - \alpha p I$ with the injected pressure inside p. That is the instant injection $Q = (p_{\text{inj}}/M)\,\delta(t)$, whose state just after $t = 0$ has $\sigma = -\alpha p_{\text{inj}} I$, not SeisSol's $\sigma = 0$.

In (4.3) the starting pressure drops out of the solid equation, because σ does not contain it. It enters the fluid equation as the fixed body force $-\nabla p_{\text{init}}$, and the fault through p. The injected volume enters both equations: for uniform αM, as the body force $-\nabla(\alpha MV)$ on the solid and $-\nabla(MV)$ on the fluid.

4.3 Conditions on the fault and the boundaries

Where	Solid	Fluid	Code status
Interior face	$[\![u]\!] = 0$, $[\![\sigma n]\!] = 0$	$[\![p]\!] = 0$, $[\![w \cdot n]\!] = 0$	DG weak continuity, §6
Fault, permeable (D3)	$[\![u]\!] \cdot n = 0$, $[\![u]\!] = d$ tangential, $[\![\sigma n]\!] = 0$, friction law	$[\![p]\!] = 0$, $[\![w \cdot n]\!] = 0$	§7
Fault, sealed (P6)	as above	$w^{\pm} \cdot n^{\pm} = 0$; p may jump	§7.5
Free surface	$\sigma n = 0$	$p = 0$ (drained)	natural, no terms
Absorbing	$\sigma n = -\rho\big(c_p(\dot{u} \cdot n)n + c_s \dot{u}_t\big)$	$p = 0$ (drained)	dashpot C (`integrator/dg_absorbing_bc_integrator.hpp`)
Dirichlet (QD loading)	$u = g$	$p = 0$ (drained), or no flow	existing SIPG Dirichlet terms, §6.6
No-flow wall (tests V-Q1–V-Q3)	$\sigma n = 0$ or $u = g$	$w \cdot n = 0$	§6.6

The free surface is drained in the total pressure, as in SeisSol, whose free-surface condition zeroes $(\sigma_{nn}, \sigma_{ns}, \sigma_{nd}, p)$. On the absorbing boundary the dashpot acts on the total traction. It should use the bulk density ρ and the undrained speed $c_p = \sqrt{(\lambda_u + 2\mu)/\rho}$, because below f_c the fast wave travels at the undrained speed (inferred). The undrained sample of §2.3 hands exactly (λ_u, μ, ρ) to the existing dashpot. A drained outer boundary reflects part of the slow wave; in `tpv5-poro` the boundaries are more than 45 km away and no wave reaches them in 2 s (plan §3.1). A poroelastic absorbing condition is later work.

4.4 Energy

Proposition 4.1. Without sources ($Q = 0$, $p_{\text{init}} = 0$), forcing, absorbing boundaries or fault, the energy

$$E = \tfrac{1}{2}\int_\Omega \big(\rho|\dot{u}|^2 + 2\rho_f\, \dot{u} \cdot \dot{w} + m|\dot{w}|^2\big) + \tfrac{1}{2}\int_\Omega \big(\varepsilon(u) : C_d : \varepsilon(u) + M\zeta^2\big) \tag{4.5}$$

satisfies $\dot{E} = -\int_\Omega b|\dot{w}|^2 \le 0$. The kinetic part is positive because $\rho m - \rho_f^2 > 0$.

Proof. Multiply (4.1) by $\dot{u}$ and (4.2) by $\dot{w}$, add, and integrate. The inertia terms give the time derivative of the kinetic part. Integration by parts gives $-\int(\nabla \cdot \sigma) \cdot \dot{u} = \int \sigma : \varepsilon(\dot{u}) = \int \varepsilon(u) : C_d : \varepsilon(\dot{u}) + \int \alpha M\zeta\, \nabla \cdot \dot{u}$ and $\int \nabla p \cdot \dot{w} = -\int p\, \nabla \cdot \dot{w} = \int M\zeta\, \nabla \cdot \dot{w}$; the boundary terms vanish on a free surface. The two $M\zeta$ terms add to $\int M\zeta\dot{\zeta}$. What remains is $\int b|\dot{w}|^2$. For the kinetic part, $\rho m - \rho_f^2 = \rho_f^2\big((1-\phi)\rho_s T/(\phi\rho_f) + T - 1\big) > 0$ because $T \ge 1$. ∎

With the fault, the friction work $\int_{\Gamma_F}(T + S) \cdot V \le 0$ is added, since V is anti-parallel to the shear traction (`domain/nitsche_wave_operator.hpp:1053-1058`). The absorbing boundary adds $-\int \rho\big(c_p(\dot{u} \cdot n)^2 + c_s|\dot{u}_t|^2\big)$.

4.5 Wave speeds

For plane waves with no drag (the high-frequency limit, which sets the time step), the P speeds solve

$$\det\left(\begin{pmatrix} \lambda_u + 2\mu & \alpha M \\ \alpha M & M \end{pmatrix} - c^2 \begin{pmatrix} \rho & \rho_f \\ \rho_f & m \end{pmatrix}\right) = 0, \qquad c_s^2 = \frac{\mu}{\rho - \rho_f^2/m}. \tag{4.6}$$

This follows from (4.1)–(4.3) with $u = Ue^{i(kx-\omega t)}e_x$, $w = We^{i(kx-\omega t)}e_x$ and $b = 0$: then $\sigma_{xx} = (\lambda_u + 2\mu)u' + \alpha M w'$ and $p = -M(\alpha u' + w')$. The larger root is the fast P speed c_{fast} used by `PoroMaxDt`; the smaller is the slow wave. At seismic frequencies ($f \ll f_c$) the fast wave travels at the undrained speed $\sqrt{(\lambda_u + 2\mu)/\rho}$. For `tpv5-poro`: $c_{\text{fast}} = 3590$ m/s, slow 1056 m/s, $c_s = 2097$ m/s, undrained P 3443 m/s and drained P 3296 m/s.

4.6 Relation to the three-field (u, w, p) form

Some papers and notes keep the pore pressure as a third unknown. The note Chunhui Zhao shared on 20 Oct 2026 (its §2.4, following Sivaselvan et al. 2016 and Mella et al. 2023) writes, in our names and with the injection rate added,

$$\rho\ddot{u} + \rho_f\ddot{w} - \nabla \cdot (\sigma' - \alpha p I) = f, \qquad \rho_f\ddot{u} + m\ddot{w} + b\dot{w} + \nabla p = 0, \qquad \frac{\dot{p}}{M} + \alpha\nabla \cdot \dot{u} + \nabla \cdot \dot{w} = Q, \tag{4.7}$$

where $\sigma' = C_d : \varepsilon(u)$ is Biot's effective stress, so $\sigma' - \alpha p I$ is the total stress.

Proposition 4.2 (the two forms are the same). Let α and M be independent of time, and start from $u = w = 0$, $p = p_{\text{init}}$ at $t = 0$. Then (u, w) solves (4.1)–(4.3) if and only if (u, w, p), with $p = p_{\text{init}} + M(V - \zeta)$, solves (4.7) with the total stress written $\sigma' - \alpha(p - p_{\text{init}})I$. For $p_{\text{init}} = 0$, the general case, the two total stresses are identical.

Proof. Differentiate $p = p_{\text{init}} + M(V - \alpha\nabla \cdot u - \nabla \cdot w)$ in time: $\dot{p}/M = Q - \alpha\nabla \cdot \dot{u} - \nabla \cdot \dot{w}$, which is the third equation of (4.7). Conversely, integrate that equation from $t = 0$ with the starting state. The first two equations are the same in both forms once σ of (4.3) is substituted. ∎

In the three-field note	Here	Remark
σ ("effective stress")	$C_d : \varepsilon(u)$	Its $\sigma - \alpha p I$ is our total stress σ (4.3).
$\rho^f \tau_t / \phi$	$m = T\rho_f/\phi$	$\tau_t = T$, the tortuosity.
$(\mu_f/\kappa)\,\dot{w}$ with $\kappa = k/\mu_f$	$b\dot{w}$, $b = \eta/\kappa$ with κ the intrinsic permeability	With the note's $\kappa = k/\mu_f$ the term should read $\dot{w}/\kappa = (\mu_f/k)\dot{w}$. As printed it is μ_f^2/k, which has the wrong units.
$\dot{p}/M + \alpha\nabla\cdot\dot{u} + \nabla\cdot\dot{w} = 0$	(4.3) with $Q = 0$	Proposition 4.2.
$\rho = (1-\phi)\rho^s + \phi\rho^f$, $\dot{w} = \phi(\dot{u}_f - \dot{u})$	The same (§4.1)	
Its eq. (31): $(1-\phi)\rho^s\ddot{u} + \phi\rho^f\ddot{u}_f = \phi\rho\ddot{u} + \phi\rho^f(\ddot{u}_f - \ddot{u})$	$\rho\ddot{u} + \rho_f\ddot{w}$ in (4.1)	Adding and subtracting $\phi\rho^f\ddot{u}$ gives $\rho\ddot{u} + \phi\rho^f(\ddot{u}_f - \ddot{u})$, which is $\rho\ddot{u} + \rho_f\ddot{w}$ by the definition of w. The factor ϕ in front of $\rho\ddot{u}$ is a typo.

What the choice of unknowns changes. Integrating the mass balance once makes p an algebraic function of (u, w). There is no pressure space and no pairing condition between a pressure space and w, and the system stays symmetric and second order in time, which is what lets the solid block reuse today's SIPG operator and the explicit Newmark step (D1, F2). The pressure is evaluated from (u, w) wherever it is needed: on the fault (7.2) and at receivers. A mixed (u, w, p) discretization is the usual choice when M is very large compared with the frame moduli (nearly incompressible fluid and grains), where the $M\zeta^2$ term acts like a penalty. Neither benchmark is in that regime: $M = 6.02$ GPa against $\lambda_u = 6.92$ GPa in `tpv5-poro`, and 20.1 against 32.0 GPa in the TPV205 set (inferred from these ratios, not tested).

Gravity and the reference state. The note's Darcy law with gravity, $q = -(k/\mu_f)\nabla(p + \rho_f g z)$ (its eq. 28), is the low-frequency limit of the second equation of (4.7) with a gravity body force. Under (A4) the background, including the hydrostatic pressure, is in equilibrium with gravity, and our p is measured from it, so the gravity term cancels. What remains is the Darcy law $b\dot{w} + \nabla p = 0$ of the quasi-static limit (§10.1). The background pressure p_0 enters only the friction law (7.4).

Two effective stresses. The note's "effective stress" is Biot's, $\sigma + \alpha p I$, which carries the deformation of the frame. The friction law uses Terzaghi's effective normal stress, $\sigma_n' = \sigma_n - p$ in compression-positive terms, with coefficient 1 (§7.3), as SeisSol does.

5. From the strong form to the weak form on one element

> **Why this step.** DG works element by element. Integrating by parts on one element produces the face terms that SIPG then replaces by numerical fluxes. The two fluxes that appear are the total traction σn (paired with the solid test v) and the pressure p (paired with the normal component of the fluid test q). Both are continuous across faces for the exact solution, which is what makes the SIPG form consistent in §6.

Multiply (4.1) by a test field v and (4.2) by a test field q, and integrate over an element K:

$$\begin{aligned} &\int_K (\rho\ddot{u} + \rho_f\ddot{w})\cdot v + (\rho_f\ddot{u} + m\ddot{w} + b\dot{w})\cdot q && \text{inertia and drag} \\ &-\int_K (\nabla\cdot\sigma)\cdot v + \int_K \nabla p\cdot q = \int_K f\cdot v. && \text{stress and pressure} \end{aligned}$$

Integrate the stress and pressure terms by parts. Because σ is symmetric, $\sigma : \nabla v = \sigma : \varepsilon(v)$, and by (4.3)

$$-\int_K (\nabla\cdot\sigma)\cdot v = \int_K \varepsilon(u) : C_d : \varepsilon(v) + \int_K \alpha M\zeta(u,w)\,\nabla\cdot v - \int_K \alpha M V\,\nabla\cdot v - \int_{\partial K} (\sigma n_K)\cdot v,$$

$$\int_K \nabla p\cdot q = \int_K M\zeta(u,w)\,\nabla\cdot q - \int_K \big(p_{\text{init}} + MV\big)\nabla\cdot q + \int_{\partial K} p\,q\cdot n_K.$$

The two $M\zeta$ volume terms combine into $\int_K M\zeta(u,w)\,\zeta(v,q)$, and the two MV terms into $-\int_K MV\,\zeta(v,q)$. The element weak form is therefore

$$\begin{aligned} &\int_K (\rho\ddot{u} + \rho_f\ddot{w})\cdot v + (\rho_f\ddot{u} + m\ddot{w} + b\dot{w})\cdot q + \int_K \varepsilon(u) : C_d : \varepsilon(v) + \int_K M\zeta(u,w)\,\zeta(v,q) \\ &-\int_{\partial K} (\sigma n_K)\cdot v + \int_{\partial K} p\,q\cdot n_K = \int_K f\cdot v + \int_K \big(MV\,\zeta(v,q) + p_{\text{init}}\,\nabla\cdot q\big). \end{aligned} \tag{5.1}$$

Black terms are today's elastic weak form. **Blue** is the new mass and drag; **orange** is new. Expanding ζ, the volume stiffness of (5.1) is

$$\varepsilon(u) : C_d : \varepsilon(v) + M\zeta(u,w)\zeta(v,q) = \underbrace{\varepsilon(u) : C_u : \varepsilon(v)}_{\text{elastic, }\lambda_u} + \alpha M\,\nabla\cdot w\,\nabla\cdot v + \alpha M\,\nabla\cdot u\,\nabla\cdot q + M\,\nabla\cdot w\,\nabla\cdot q,$$

and the part of the total stress that depends on the unknowns is $\sigma(u,w) = C_u : \varepsilon(u) + \alpha M(\nabla\cdot w)I$; the rest, $-\alpha MVI$, is a load. So the solid part is exactly the elastic operator with λ_u, and everything new is built from divergences.

Term of (5.1)	Becomes	Where it goes in code
Inertia and drag	$\mathbb{M}\ddot{x} + \mathbb{D}\dot{x}$, eq. (8.2)	`PoroMass`; drag force in `PoroNitscheOperator`
$\int_K \varepsilon : C_u : \varepsilon$ and the $C_u\varepsilon(u)n$ part of $-\int_{\partial K} \sigma n_K \cdot v$	a_E^u in (6.3)	`ElasticityDomainOperator` with λ_u, unchanged
αM volume terms, the $\alpha M \nabla \cdot w\, n$ part of σn, the $\alpha M \nabla \cdot u$ part of p	$B_{\alpha M}$ in (6.3)	`DGPoroDivIntegrator`, coupling blocks
$M \nabla \cdot w \, \nabla \cdot q$ and the $M \nabla \cdot w$ part of p	B_M plus penalty in (6.3)	`DGPoroDivIntegrator`, fluid block
The V and p_{init} terms, and their parts of σn and p on faces	source loads, eq. (6.6)	`AssembleSourceRHS`

6. From the element weak form to SIPG

6.1 Summing over elements: what the exact solution says on a face

Sum (5.1) over all elements and apply Lemma 3.1 to both boundary terms. On an interior face they become

$$-\{\{\sigma n\}\} \cdot [\![v]\!] - [\![\sigma n]\!] \cdot \{\{v\}\} + \{\{p\}\}[\![q \cdot n]\!] + [\![p]\!]\{\{q\}\} \cdot n.$$

For the exact solution the total traction and the pressure are continuous, $[\![\sigma n]\!] = 0$ and $[\![p]\!] = 0$, on interior faces and on the permeable fault. Two terms drop, and the exact face term is

$$-\sigma n \cdot [\![v]\!] + p\,[\![q \cdot n]\!]. \tag{6.1}$$

This holds whatever jumps the material has (A3), because no material constant was pulled out of a flux.

Proposition 6.1 (finding F2). Suppose instead the pressure part of the stress is moved into the fluid flux: write $\sigma = \sigma_d - \alpha p' I$ with $\sigma_d = C_d : \varepsilon$ and $p' = p - p_{\text{init}} - MV$, and pair p' with $(\alpha v + q) \cdot n$, as plan revision 2.2 does. Then the dropped term $[\![\sigma_d n]\!] \cdot \{\{v\}\} = [\![\alpha]\!]\, p' \,\{\{v\}\} \cdot n$ is not zero where α jumps, and the plan's penalized jump $[\![(\alpha u + w) \cdot n]\!] = [\![\alpha]\!]\, u \cdot n$ is not zero for the exact solution. Both are consistent only when α is continuous. Patch test U2-6 of plan revision 3 checks this with an exact piecewise-linear solution across an α jump.

Proof. Total traction is continuous, so $[\![\sigma_d n]\!] = [\![\sigma n + \alpha p' n]\!] = [\![\alpha]\!] p' n$ with p' continuous. The exact solution has $[\![u]\!] \cdot n = 0$ and $[\![w \cdot n]\!] = 0$, so $[\![(\alpha u + w) \cdot n]\!] = [\![\alpha u]\!] \cdot n = [\![\alpha]\!]\, u \cdot n$ by the product rule. ∎

6.2 The SIPG form

Why this step. The exact face term (6.1) needs face values for σn and p. SIPG uses averages of the discrete fluxes, adds the symmetric counterparts so that the form stays symmetric (the energy argument needs this), and adds penalties for coercivity. The fluid needs only $H(\mathrm{div})$ regularity, so only the normal jump of w is penalized; tangential jumps of w are allowed. Antonietti et al. (2022) also penalize the full jump of u and only the normal jump of w (review eqs. 32–36, checked on 20 Oct 2026). They apply the fluid form to $\beta u + w$ (review eq. 32), which is the plan revision 2.2 form that Proposition 6.1 shows needs a continuous α.

Antonietti, Botti, Mazzieri, Nati Poltri (2022), SIAM J. Sci. Comput. 44(1) B1–B28; Antonietti, Botti, Mazzieri (2022), Vietnam J. Math. 50(4) 997–1028.

Take the parts of the total stress and the pressure that depend on the unknowns, $\sigma(u,w) = C_d : \varepsilon(u) + \alpha M \zeta(u,w) I$ and $-M\zeta(u,w)$. Replace σn and p on faces by their averages, move the source parts ($-\alpha M V I$ in the stress, $p_{\text{init}} + MV$ in the pressure) to the right side, and add the symmetry and penalty terms:

$$\begin{aligned} a_h\big((u,w),(v,q)\big) = &\sum_K \int_K \Big(\varepsilon(u) : C_d : \varepsilon(v) + M\zeta(u,w)\,\zeta(v,q)\Big) && \text{volume} \\ &- \sum_F \int_F \Big(\{\{\sigma(u,w)n\}\} \cdot [\![v]\!] + \{\{\sigma(v,q)n\}\} \cdot [\![u]\!]\Big) && \text{solid flux} \\ &- \sum_F \int_F \Big(\{\{M\zeta(u,w)\}\}[\![q \cdot n]\!] + \{\{M\zeta(v,q)\}\}[\![w \cdot n]\!]\Big) && \text{fluid flux} \\ &+ \sum_F \int_F \Big(\sigma_u\, [\![u]\!] \cdot [\![v]\!] + \sigma_w\, [\![w \cdot n]\!][\![q \cdot n]\!]\Big). && \text{penalties} \end{aligned} \tag{6.2}$$

The face sums run over interior faces and permeable fault faces; on the fault $[\![u]\!]$ becomes $[\![u]\!] - d$ in the symmetry and penalty terms (§7.1). Boundary faces are in §6.6. Expanding σ and ζ as at the end of §5 splits (6.2) into blocks:

$$\begin{aligned} a_h\big((u,w),(v,q)\big) &= a_E^u(u,v) + B_{\alpha M}(w,v) + B_{\alpha M}(u,q) + B_M(w,q) + \sum_F \int_F \sigma_w [\![w \cdot n]\!][\![q \cdot n]\!], \\ B_c(a,b) &= \sum_K \int_K c\,(\nabla \cdot a)(\nabla \cdot b) - \sum_F \int_F \Big(\{\{c\nabla \cdot a\}\}[\![b \cdot n]\!] + \{\{c\nabla \cdot b\}\}[\![a \cdot n]\!]\Big), \end{aligned} \tag{6.3}$$

where a_E^u is today's elastic SIPG form (§6.3) with C_u (that is, λ_u) and the penalty σ_u. $B_c(a,b)$ is symmetric in a,b, so (6.3) is symmetric. In matrix terms, $Z_{uw} = Z_{wu}^{\mathsf{T}}$ is the matrix of $B_{\alpha M}$ and Z_{ww} the matrix of B_M plus the σ_w penalty.

> *Derivation of (6.3).* $\sigma(u,w)n\cdot[\![v]\!] = C_u\varepsilon(u)n\cdot[\![v]\!] + \alpha M\,\nabla\cdot w\,[\![v\cdot n]\!]$, because $\alpha M\zeta = \alpha^2 M\nabla\cdot u + \alpha M\nabla\cdot w$ and $C_d\varepsilon + \alpha^2 M(\nabla\cdot u)I = C_u\varepsilon$. Likewise $M\zeta(u,w)[\![q\cdot n]\!] = \alpha M\,\nabla\cdot u\,[\![q\cdot n]\!] + M\,\nabla\cdot w\,[\![q\cdot n]\!]$. Averages are linear, so each face term splits the same way. Collecting the terms with (trial, test) = (u,v), (w,v), (u,q), (w,q) gives the four blocks. ∎

> **Proposition 6.2 (locked fluid).** With $w = q = 0$, a_h is today's elastic SIPG form with (λ_u,μ), term by term, including its penalty, its Dirichlet terms and its slip load. So a poroelastic run whose w stays zero reproduces today's elastic operator built from the undrained sample bit for bit, which is a cheap first test of the implementation.

6.3 The solid block: today's SIPG with the undrained modulus

For a stiffness C with Lamé moduli (λ,μ), today's form is

$$a_E^C(u,v) = \sum_K \int_K \varepsilon(u) : C : \varepsilon(v) - \sum_F \int_F \Big(\{\{C\varepsilon(u)n\}\}\cdot[\![v]\!] + \{\{C\varepsilon(v)n\}\}\cdot[\![u]\!] \Big) + \sum_F \int_F \sigma_F\,[\![u]\!]\cdot[\![v]\!],$$

and $a_E^u = a_E^{C_u}$. This is what the code assembles. The face block of `DGElasticityIPCombinedIntegrator` has the coefficients below, with $\epsilon = -1$ (SIPG, `domain/elasticity_operator.hpp:468`).

Block (test, trial)	c_0 (consistency)	c_1 (symmetry)	c_2 (penalty)	Code
(+, +)	$-\frac{1}{2}$	$\frac{1}{2}\epsilon$	$+\sigma_F$	`…combined_integrator.hpp:129-139`
(+, −)	$-\frac{1}{2}$	$-\frac{1}{2}\epsilon$	$-\sigma_F$	`:144-154`
(−, +)	$+\frac{1}{2}$	$\frac{1}{2}\epsilon$	$-\sigma_F$	`:157-167`
(−, −)	$+\frac{1}{2}$	$-\frac{1}{2}\epsilon$	$+\sigma_F$	`:170-180`
Dirichlet	-1	ϵ	$+\sigma_F$	`:129-139`

Summing the four interior blocks gives exactly the three face terms of a_E^C. The code then averages each face matrix with its transpose (`:183-195`), which changes nothing for SIPG. The penalty is

$$\sigma_u = \frac{\kappa_F}{4}\sum_{i=\pm}(D+1)\,\frac{p(p+D-1)}{D}\,\frac{D\,|J_F|}{|J_{K_i}|}\,\frac{(D\lambda_{u,i}+2\mu_i)^2}{2\mu_i}, \qquad D=3, \tag{6.4}$$

with κ_F the `penalty_factor`; a boundary face uses its one side's term without the $\frac{1}{4}$ (`ComputePenalty`, `…combined_integrator.hpp:857-881`). This is today's σ_F evaluated with λ_u.

6.4 The divergence blocks and the fluid penalty

The new form B_c of (6.3) is a SIPG form for the divergence: volume coefficient c, flux $\{\{c\nabla\cdot a\}\}$, conjugate jump $[\![a\cdot n]\!]$. It is used three times: $c = \alpha M$ for the two coupling blocks (no penalty) and $c = M$ for the fluid block (with penalty). On one element, with the divergence matrix B_e ($B_e a_e$ = the divergence of a at the quadrature points) and the quadrature weights W, the volume part is $c\,B_e^{\mathsf{T}} W B_e$, so the element matrix of the new blocks on (u,w) is

$$\begin{pmatrix} 0 & \alpha M\,B_e^{\mathsf{T}} W B_e \\ \alpha M\,B_e^{\mathsf{T}} W B_e & M\,B_e^{\mathsf{T}} W B_e \end{pmatrix},$$

next to today's elastic element matrix with λ_u in the uu position. The face blocks follow the pseudocode of §2.3. The proposed fluid penalty mirrors (6.4) with $(D\lambda+2\mu)^2/(2\mu)$ replaced by M:

$$\sigma_w = \frac{\kappa_w}{4}\sum_{i=\pm}(D+1)\,\frac{p(p+D-1)}{D}\,\frac{D\,|J_F|}{|J_{K_i}|}\,M_i. \tag{6.5}$$

The reason: in (6.4) the flux coefficient enters squared and the coercivity constant divides; for the fluid flux $M\zeta$ both are M, which gives $M^2/M = M$ (inferred by analogy with the inverse-trace argument). The solid flux now also contains $\alpha M\zeta$, which σ_u must control; it does, since $(D\lambda_u+2\mu)^2/(2\mu) \geq D\lambda_u + 2\mu \geq \alpha^2 M$ (inferred, same argument). κ_w starts at 1. Test U2-5 of plan revision 3 measures coercivity and test U2-18 the effect on the time step.

6.5 The sources

Moving the source parts of the face terms $\{\{\sigma n\}\}\cdot[\![v]\!]$ and $\{\{p\}\}[\![q\cdot n]\!]$ to the right side, together with the volume terms of (5.1), gives the source load

$$F_{\mathrm{src}}(t;v,q) = \underbrace{\sum_K \int_K MV\,\zeta(v,q) - \sum_F \int_F \Big(\{\{\alpha MV\}\}[\![v\cdot n]\!] + \{\{MV\}\}[\![q\cdot n]\!] \Big)}_{F_V(t)\text{: solid and fluid rows}} + \underbrace{\sum_K \int_K p_{\mathrm{init}}\,\nabla\cdot q - \sum_F \int_F \{\{p_{\mathrm{init}}\}\}[\![q\cdot n]\!]}_{F_{\mathrm{init}}\text{: fluid rows}}. \tag{6.6}$$

Both parts have the pattern of the consistency half of B_c, with a known scalar field in place of $\nabla\cdot a$, so one routine assembles them: V with the coefficient αM on the solid rows and M on the fluid rows, and p_{init} with the coefficient 1 on the fluid rows. The face sums run over interior and permeable fault faces. On a drained boundary the total pressure is zero, so the boundary term $\int p\,q\cdot n$ vanishes and no boundary datum is needed.

F_V changes with time and is assembled at every step. `PoroSource` keeps V at the quadrature points and updates it by the trapezoid rule, $V^{n+1} = V^n + \frac{\Delta t}{2}(Q^n + Q^{n+1})$, or exactly when Q has a closed-form time integral. A well inside an element, $Q = Q_w(t)\,\delta(x - x_w)$, gives $MV_w(t)\,\zeta(v,q)(x_w)$ and no face terms. For the comparison with SeisSol, $V = 0$ and only F_{init} remains. For a continuous p_{init} it is the weak form of the body force $-\nabla p_{\text{init}}$ of §4.2.

6.6 Boundary faces

From the boundary part of Lemma 3.1, a boundary face contributes $-\int_F (\sigma n)\cdot v + \int_F p\,q\cdot n$. Each condition of §4.3 fixes these terms as follows.

Boundary	Solid terms	Fluid terms
Free surface ($\sigma n = 0$, $p = 0$)	none (natural)	none (natural)
Absorbing (dashpot, $p = 0$)	$c(\dot u, v)$: existing `DGAbsorbingBCIntegrator` with (λ_u, μ, ρ)	none
Dirichlet $u = g$, drained	existing one-sided SIPG Dirichlet terms with λ_u, plus the coupling terms $-\int \alpha M(\nabla\cdot w)(v\cdot n) - \int \alpha M(\nabla\cdot q)\big((u-g)\cdot n\big)$; load $-\int \alpha M V(v\cdot n)$	none
Traction-free, no-flow ($w\cdot n = 0$)	none	$-\int M\zeta(u,w)\,q\cdot n - \int M\zeta(v,q)\,w\cdot n + \int \sigma_w (w\cdot n)(q\cdot n)$; load $-\int (p_{\text{init}} + MV)\,q\cdot n$
Dirichlet $u = g$, no-flow	as Dirichlet, drained	as traction-free, no-flow

The Dirichlet coupling terms come from the $\alpha M(\nabla\cdot w)I$ part of the total stress in the one-sided consistency and symmetry terms; without them the Dirichlet face would be inconsistent once $w \neq 0$. Tests V-Q1–V-Q3 (Terzaghi, Mandel, Cryer) need the no-flow rows and also rollers (normal displacement fixed, shear traction free). Whether the elastic code supports rollers is checked at the start of P7 (not yet verified).

6.7 The full semi-discrete problem

> **Problem 6.3.** Find $(u,w)(t) \in V_h \times V_h$ such that for all $(v,q) \in V_h \times V_h$
>
> $$m_h\big((\ddot u, \ddot w, \dot w), (v,q)\big) + c(\dot u, v) + a_h\big((u,w),(v,q)\big) = \ell_E(t;v) + \ell_{\text{slip}}(d;v) + F_{\text{src}}(t;v,q), \tag{6.7}$$
>
> with $m_h = \sum_K \int_K (\rho\ddot u + \rho_f \ddot w)\cdot v + (\rho_f \ddot u + m\ddot w + b\dot w)\cdot q$, c the absorbing dashpot, ℓ_E today's Dirichlet and body loads, ℓ_{slip} the slip load (7.1), and F_{src} the source load (6.6).

> **Proposition 6.4 (consistency and energy).** (i) The exact solution, if smooth on each element, satisfies (6.7), for any jumps of the material between elements. (ii) With no loads, absorbing faces or fault, the discrete energy $E_h = \frac12 m_h^{\text{kin}} + \frac12 a_h\big((u,w),(u,w)\big)$ satisfies $\dot E_h = -\int b|\dot w|^2$.

Proof. (i) The sum of (5.1) over elements is (6.7) with the face terms (6.1); the added symmetry and penalty terms vanish because $[\![u]\!] - d = 0$ and $[\![w\cdot n]\!] = 0$. (ii) Test with $(v,q) = (\dot u, \dot w)$. a_h is symmetric, so its contribution is $\frac{d}{dt}\frac12 a_h(\cdot,\cdot)$, and the inertia gives the kinetic part as in Proposition 4.1. ∎

E_h is an energy (positive) when the penalties make a_h coercive in the energy norm $\int \varepsilon : C_d : \varepsilon + M\zeta^2$ plus the jump terms. For (6.4)–(6.5) this is inferred, not proven here; test U2-5 checks it numerically (plan revision 3, §6).

7. The fault

7.1 Slip enters through the solid block only

Formulation S keeps slip d as a prescribed field in the face terms: on fault faces, $[\![u]\!]$ in the symmetry and penalty terms of a_E^u becomes $[\![u]\!] - d$. Moving the d terms to the right side gives the slip load

$$\ell_{\text{slip}}(d;v) = \sum_{F\subset\Gamma_F} \int_F \Big(\sigma_u\, d\cdot [\![v]\!] - \{\{C_u \varepsilon(v) n\}\}\cdot d\Big) = (G^u d)\cdot v. \tag{7.1}$$

This is what `AssembleSlipFaceRHS` assembles: on both sides the symmetry coefficient is $c_1 = \epsilon/2$, with $+\sigma_F$ on the $+$ side and $-\sigma_F$ on the $-$ side (`…combined_integrator.hpp:274-325`), here with λ_u. The step uses it through `SlipLoad` or `AddSlipLoadFast` (`domain/nitsche_wave_operator.hpp:470, 858-872`). The coupling block $B_{\alpha M}(u,q)$ also contains $[\![u\cdot n]\!]$, but slip is tangential ($d\cdot n = 0$), so it carries no slip and needs no lifting.

7.2 The fault pressure

> **Why this step.** The fault must see the pressure of the same numerical flux that the bulk uses. A plain average of the two sides' pressures would repeat, for the pressure, the instability that the code warns against for the traction

(`domain/nitsche_wave_operator.hpp:49-51`, citing Uphoff et al. §5.1).

Collect every face term of (6.2) and (6.6) that multiplies the test's normal jump $[\![q \cdot n]\!]$: $-\{\{M\zeta(u,w)\}\} + \sigma_w [\![w \cdot n]\!] + \{\{p_{\text{init}} + MV\}\}$ (the last one moved back from the load). With the discrete pressure $p_h = p_{\text{init}} + MV - M\zeta(u,w)$, this is the numerical face pressure

$$\hat{p} = \{\{p_h\}\} + \sigma_w [\![w \cdot n]\!], \tag{7.2}$$

the discrete counterpart of p in the exact face term (6.1). Both $\{\{p_h\}\}$ and $[\![w \cdot n]\!]$ are unchanged when the two sides swap, so $\hat{p}$ needs none of the normal-sign bookkeeping the traction needs (`ProjectTractionToFaultDOFs`, `…combined_integrator.hpp:715`). Sign check: if more fluid flows into the face from the $+$ side than leaves on the $-$ side, $[\![w \cdot n]\!] > 0$ and the pressure rises.

7.3 Total traction and effective normal stress

Collect the face terms that multiply $[\![v]\!]$, with the source part $\{\{\alpha MV\}\}[\![v \cdot n]\!]$ moved back from the load: $-\big(\{\{\sigma(u,w)n\}\} - \{\{\alpha MV\}\}n - \sigma_u([\![u]\!] - d)\big) \cdot [\![v]\!] = -T \cdot [\![v]\!]$. With $\sigma(u,w) = C_u\varepsilon(u) + \alpha M(\nabla \cdot w)I$ the total traction is

$$T = \underbrace{\{\{C_u\varepsilon(u)n\}\} - \sigma_u([\![u]\!] - d)}_{T_h\text{: today's kernel with } \lambda_u} + \{\{\alpha M\,(\nabla \cdot w - V)\}\}\,n. \tag{7.3}$$

T_h is exactly what `ComputeTractionAtQuadPoints` returns (`…combined_integrator.hpp:514-552`) when the operator is built from the undrained sample. The second term is new (`CouplingTraction`). With prestress S (total stress, A4) and background pressure p_0, Terzaghi's rule with coefficient 1 gives

$$\sigma_n' = -(T + S) \cdot n - (p_0 + \hat{p}). \tag{7.4}$$

This matches SeisSol, which uses $\sigma_{n0} + \sigma_n + p_0 + p^*$ in its tension-positive convention, that is $-\sigma_n' = (T + S) \cdot n + p_0 + \hat{p}$ (`LinearSlipWeakening.h`, `FrictionSolverCommon.h`); SeisSol's fault pressure p^* comes from a 4×4 impedance solve instead of (7.2). In code, hook H2 adds the coupling traction to the trial traction before box 8, and passes $p_0 + \hat{p}$ to `CloseDOF`, which subtracts it from σ_n before the range check at line 495 (`domain/nitsche_wave_operator.hpp:488-516`).

> **Proposition 7.1 (the closure is unchanged).** $\hat{p}$ and the coupling traction depend only on $(u,w)^{n+1}$, which step 1 fixes before the closure, and on the source at t_{n+1}; and slip enters T only through the tangential term $\sigma_u d$. So $\partial\sigma_n'/\partial V^{n+1} = 0$, and the within-step Jacobian stays $\gamma = \sigma_u \Delta t/2$ (plus the Kelvin–Voigt slope, as now), computed by today's `ClosureGamma` with λ_u.

7.4 Prestress and the sources on the fault

S is the background total traction on the fault, as today. p_0 is the background pore pressure; in P0 it is the static P_p of the deck (`tpv205/configs/tpv205_nitsche.toml:57-59`), replacing today's abort (`drivers/nitsche_dyn_driver.cpp:880-882`). `tpv5-poro` has $p_0 = 0$ (`fault.yaml`).

Fluid injected near the fault first acts undrained. Before the rock deforms and the fluid flows, V raises $\hat{p}$ by MV and makes $T \cdot n$ more compressive by αMV, so σ_n' drops by only $(1-\alpha)MV$. After that, deformation and flow change both terms, and (7.4) follows them.

A starting pressure is part of $\{\{p_h\}\}$ and absent from T, so at $t = 0$ the fault sees $\sigma_n' = -S \cdot n - p_0 - p_{\text{init}}$. At the `tpv5-poro` hypocentre that is $120 - 45.2 = 74.8$ MPa: a static strength of $0.677 \times 74.8 = 50.6$ MPa against 70 MPa of shear stress. So rupture starts at once wherever $p_{\text{init}} > 16.6$ MPa on the fault, a patch of radius 3.2 km. This is how `tpv5-poro` nucleates, and why the comparison with SeisSol needs the starting pressure rather than an instant injection (finding F1).

7.5 The sealed fault (P6)

For a sealed fault, the fault faces are removed from the fluid face sums of (6.2), and each side gets the no-flow row of §6.6 (one-sided, $w^{\pm} \cdot n^{\pm} = 0$). Each side then has its own pressure, $p^{\pm} = p_h^{\pm} - \sigma_w^{\pm} w^{\pm} \cdot n^{\pm}$, and friction needs a rule for which pressure it sees (plan §3.5). That choice is left to P6.

8. Matrix form

8.1 Vectors and blocks

Order the unknowns as $x = [u; w]$, each in the existing vector DG layout (`Ordering::byNODES`, `domain/nitsche_mass.hpp:21-22`). Problem 6.3 becomes

$$\mathbb{M}\ddot{x} + \mathbb{D}\dot{x} + \mathbb{K}x = F(t,d), \qquad \mathbb{K} = \begin{pmatrix} K_E^u & Z_{uw} \\ Z_{wu} & Z_{ww} \end{pmatrix}, \qquad F = \begin{pmatrix} F_E(t) + G^u d + F_{V,u}(t) \\ F_{V,w}(t) + F_{\text{init}} \end{pmatrix}. \tag{8.1}$$

K_E^u is today's stiffness (`ApplyStiffness`) built with λ_u, F_E today's Dirichlet load (`AssembleDirichletOnlyRHS`) $G^u d$ the slip load, and F_V, F_{init} the two parts of the source load (6.6), split into solid and fluid rows. $\mathbb{K}$ is symmetric. Kelvin–Voigt damping, when on, adds $-(K_\eta \dot{u} - G_\eta V)$ to the u rows exactly as now (`domain/nitsche_wave_operator.hpp:878-907`).

8.2 Mass and drag are element-local

With the scalar DG mass M_e of `NitscheMass` and the per-element 2×2 matrices

$$M_\rho = \begin{pmatrix} \rho & \rho_f \\ \rho_f & m \end{pmatrix}, \qquad D_\rho = \begin{pmatrix} 0 & 0 \\ 0 & b \end{pmatrix}, \qquad \mathbb{M}\big|_e = M_\rho \otimes M_e \otimes I_3, \quad \mathbb{D}\big|_e = D_\rho \otimes M_e \otimes I_3 + C_e. \tag{8.2}$$

The step of §9 inverts $\mathbb{M} + \Delta t\, \mathbb{D}\Gamma$. Away from absorbing faces ($C_e = 0$), $D_\rho \Gamma = \mathrm{diag}(0, b\gamma_w)$ for both choices of Γ in (9.5), so

$$(\mathbb{M} + \Delta t\, \mathbb{D}\Gamma)^{-1}\big|_e = \big(M_\rho + \Delta t\, \gamma_w D_\rho\big)^{-1} \otimes M_e^{-1} \otimes I_3. \tag{8.3}$$

This reuses `NitscheMass`'s M_e^{-1} and adds one 2×2 inverse per element. In elements with an absorbing face the block is dense ($6n_e \times 6n_e$) and is inverted once, as `NitscheMass` already does for $\rho M_e \otimes I_3 + \frac{\Delta t}{2} C_e$ (`domain/nitsche_mass.hpp:24-30, 100-131`). The relaxation rate of the relative motion is the nonzero eigenvalue of $M_\rho^{-1} D_\rho$:

$$\frac{1}{\tau} = \frac{b\,\rho}{\rho m - \rho_f^2}. \tag{8.4}$$

9. Time stepping

9.1 Newmark with a velocity coefficient per unknown

Newmark with $\beta = 0$ and a velocity-coefficient matrix Γ:

$$x^{n+1} = x^n + \Delta t\, v^n + \tfrac{\Delta t^2}{2} a^n, \qquad v^{n+1} = v^n + \Delta t\big((I - \Gamma) a^n + \Gamma a^{n+1}\big). \tag{9.1}$$

Imposing (8.1) at t_{n+1} and writing $v^* = v^n + \Delta t (I - \Gamma) a^n$, so that $v^{n+1} = v^* + \Delta t\, \Gamma a^{n+1}$, gives the acceleration

$$(\mathbb{M} + \Delta t\, \mathbb{D}\Gamma)\, a^{n+1} = F(t_{n+1}, d^{n+1}) - \mathbb{K} x^{n+1} - \mathbb{D} v^*. \tag{9.2}$$

With $\Gamma = \frac{1}{2} I$ this is today's step: `CentralDifferenceSolver` with the damped mass $M + \frac{\Delta t}{2} C$ (`domain/nitsche_wave_operator.hpp:9-24, 278-316`). The fault advance sits between the two halves exactly as now: it needs x^{n+1} from (9.1) and returns d^{n+1} for (9.2). The source load in $F(t_{n+1})$ uses V^{n+1} (§6.5).

9.2 The scalar model and the fitted coefficient

Proposition 9.1. For $m_s \dot v + b v = f$ with constant f and a consistent start ($m_s a^0 + b v^0 = f$), the scheme (9.1)–(9.2) with coefficient γ gives $v^{n+1} - v_\infty = R(z)(v^n - v_\infty)$, with $v_\infty = f/b$, $z = \Delta t\, b / m_s$ and

$$R(z) = \frac{1 - (1 - \gamma) z}{1 + \gamma z}. \tag{9.3}$$

$R(z) = e^{-z}$, the exact decay over one step, if and only if

$$\gamma = \gamma_w(z) = \frac{1}{1 - e^{-z}} - \frac{1}{z} \in \big[\tfrac{1}{2}, 1\big), \qquad \gamma_w = \tfrac{1}{2} + \tfrac{z}{12} - \tfrac{z^3}{720} + O(z^5). \tag{9.4}$$

Proof. The equation holds at every step, so $a^k = -b\, e^k / m_s$ with $e^k = v^k - v_\infty$. Then (9.1) reads $e^{n+1} = e^n - z\big((1-\gamma) e^n + \gamma e^{n+1}\big)$, which is (9.3). Setting $R = e^{-z}$: $1 - z + \gamma z = e^{-z}(1 + \gamma z)$, so $\gamma z (1 - e^{-z}) = z - 1 + e^{-z}$, which is (9.4). The series follows from $1/(1 - e^{-z}) = 1/z + \tfrac{1}{2} + z/12 - z^3/720 + \dots$ ∎

Central difference ($\gamma = \frac{1}{2}$) gives $R \to -1$ as $z \to \infty$: the error in the fluid velocity flips sign every step instead of decaying. In `tpv5-poro`, $z \approx 2.6 \times 10^3$ at 200 m and $p = 2$ (plan §3.3). The fitted γ_w is second order where the drag is resolved ($\gamma_w - \frac{1}{2} = O(z)$) and exact for the relaxation where it is stiff. `FittedGamma` uses the series for $z < 10^{-2}$ to avoid cancellation. The numerical check of these statements is the table in plan revision 2.2, §8.4 (`drag2.py`); test U2-13 of plan revision 3 checks it in the code.

9.3 The coupled 2×2 element system, and finding F3

On one element without stiffness, (8.2) is $\rho a_u + \rho_f a_w = f_u$, $\rho_f a_u + m a_w + b v_w = f_w$. Eliminating a_u gives

$$m_{\mathrm{eff}}\, a_w + b\, v_w = f_w - \tfrac{\rho_f}{\rho} f_u, \qquad m_{\mathrm{eff}} = m - \rho_f^2/\rho, \qquad \tau = m_{\mathrm{eff}}/b,$$

which is the scalar model with $z = \Delta t/\tau$, eq. (8.4). So the w row of Γ must be $[0,\ \gamma_w(z)]$, and then the fluid relaxation is exact. The u row is a choice:

$$\Gamma_{\mathrm{diag}} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \gamma_w \end{pmatrix} \ \text{(plan)}, \qquad \Gamma_{\mathrm{mom}} = \begin{pmatrix} \frac{1}{2} & \frac{\rho_f}{\rho}\big(\frac{1}{2} - \gamma_w\big) \\ 0 & \gamma_w \end{pmatrix} \ \text{(F3)}. \tag{9.5}$$

Proposition 9.2. Over one step, the discrete total momentum $\rho v_u + \rho_f v_w$ changes by

$$\Delta t\, \tfrac{1}{2}\big(f^n + f^{n+1}\big) + \rho_f \Delta t \big(\gamma_w - \tfrac{1}{2}\big)\big(a_w^{n+1} - a_w^n\big) \ \text{ with } \Gamma_{\mathrm{diag}}, \qquad \Delta t\, \tfrac{1}{2}\big(f^n + f^{n+1}\big) \ \text{ with } \Gamma_{\mathrm{mom}},$$

where f is the total force on the momentum row (here f_u). The acceleration and the fluid velocity are the same for both choices.

Proof. By (9.1), the u row of Γ_{diag} gives $\rho \Delta v_u = \frac{\Delta t}{2} \rho (a_u^n + a_u^{n+1})$, and the w row gives $\rho_f \Delta v_w = \rho_f \Delta t \big((1 - \gamma_w) a_w^n + \gamma_w a_w^{n+1}\big)$. Adding, and using $\rho a_u + \rho_f a_w = f$ at both steps, gives the first expression. With Γ_{mom}, $\rho \Delta v_u$ gains $\rho_f \Delta t (\frac{1}{2} - \gamma_w)(a_w^{n+1} - a_w^n)$,

which cancels the extra term. The acceleration solve (9.2) involves Γ only through $\mathbb{D}\Gamma$ and $\mathbb{D}v^*$, and D_ρ has only its ww entry, so the u row of Γ does not enter. ∎

Proposition 9.3 (F3 costs nothing). Wherever $C_e = 0$, (8.3) holds for Γ_{mom} unchanged. In elements with an absorbing face, Γ_{mom} adds a coupling block $\Delta t \frac{\rho_f}{\rho}(\frac{1}{2} - \gamma_w) C_e$ inside the dense block that is inverted once anyway. Only the velocity update (9.1) changes: two extra multiply-adds per DOF.

When the drag is resolved, $\gamma_w - \frac{1}{2} = O(z)$ and the extra term of Γ_{diag} is $O(\Delta t^3/\tau)$ per step. When it is stiff, the term is nonzero only in steps where a_w changes, for example when the pressure gradient changes as a wave passes. Whether that matters in practice is not yet measured. F3 removes the question at no cost; the stability of the non-diagonal Γ is checked by test U2-18 (not proven here).

9.4 Start, stability and the time-step bound

The driver starts with $a^0 = 0$, as now. With a free-fluid start $a^0 = M_\rho^{-1} f$, the first w increment would be $z/2$ times the Darcy value $v_\infty \Delta t$ (plan revision 2.2, §8.4 table). With $a^0 = 0$ it is zero, and the first velocity is $\gamma_w z/(1 + \gamma_w z)\, v_\infty \approx v_\infty$ for large z (from the scheme; test U2-13 checks it).

A coefficient $\gamma_w > \frac{1}{2}$ lowers the undamped Newmark bound from $\omega\Delta t \le 2$ to $\omega\Delta t \le \sqrt{2/\gamma_w}$. That applies only where z is large, where the w modes are overdamped. The step therefore keeps today's formula with the fast P speed of (4.6):

$$\Delta t = \text{cfl} \cdot \min_e \frac{h_e}{p^2\, c_{\text{fast},e}}. \tag{9.6}$$

This is `ComputeMaxDt` (`solver/nitsche_cfl.hpp:68-92`) with c_{fast} in place of $\sqrt{(\lambda + 2\mu)/\rho}$. In `tpv5-poro` it is 4.1 % smaller than with the undrained P speed and 8.2 % smaller than with the drained one. Test U2-18 measures whether the penalties lower the stable constant.

9.5 How SeisSol steps the drag

SeisSol does not use Newmark or a coefficient like γ_w. It is ADER-DG: in each element a local space–time predictor solves the 13-variable system over the whole step with the drag source term treated implicitly, and an explicit corrector then updates the element (Wolf et al. 2022; the poroelastic build compiles the space–time predictor kernels, `SEISSOL_KERNELS_STP`). The drag entries of its source matrix are $e_1 = \beta_1 \nu/(\rho_1 \kappa)$ on the solid-velocity columns and $e_2 = \beta_2 \nu/(\rho_2 \kappa)$ on the relative-velocity columns (`poroelastic_Setup.h:176-189`, local copy in `tpv5-poro/seissol_reference/`). The fitted γ_w is ours (Proposition 9.1). It is the exponential-fitting idea: choose the implicit weight so that one step reproduces the exact decay e^{-z}. The two schemes share what matters for stiffness: the drag is implicit and element-local, so neither needs a global solve.

`tpv5-poro`, CFL 0.5	1000 m, p = 1	200 m, p = 1	200 m, p = 2	200 m, p = 3	100 m, p = 2	100 m, p = 3
$z = \Delta t/\tau$	5.1×10^4	1.0×10^4	2.6×10^3	1.1×10^3	1.3×10^3	567
$\gamma_w(z)$	0.99998	0.99990	0.99961	0.99912	0.99922	0.99824

The values come from `poro_test_refs.py` (plan revision 3, §6.2). In every case $\gamma_w \ge 0.998$, so in `tpv5-poro` the fluid velocity is stepped by what is nearly backward Euler, and in both codes the relative fluid velocity reaches its Darcy value within one step. The drag time scheme should therefore not separate us from SeisSol in this benchmark (inferred; the SeisSol comparisons S1 and S2 of the plan check it). The fitting matters where z is near 1, which is why the plane-wave test V-B1 of the plan also runs a permeable set ($\kappa = 10^{-7}$ m², $z \approx 0.01$). SeisSol's published `tpv5-poro` run uses clustered local time stepping (`ClusteredLTS = 2` in `parameters.par`); the comparison runs keep that setting.

10. The quasi-static limit, the quasi-dynamic step and the switch

10.1 Quasi-static Biot

Dropping inertia in (4.1)–(4.2) leaves $-\nabla \cdot \sigma = f$ and $b\dot{w} + \nabla p = 0$ (Darcy's law). The weak form is (6.7) without m_h except the drag:

$$a_h\big((u,w),(v,q)\big) + \sum_K \int_K b\, \dot{w} \cdot q = \ell(v,q). \tag{10.1}$$

In matrix form, with $B_w = b\, M_e \otimes I_3$ (element-local) $f(d)$ the slip, boundary, initial-stress and solid-row source loads, and f_w the fluid-row source loads of (6.6):

$$\begin{pmatrix} K_E^u & Z_{uw} \\ Z_{wu} & Z_{ww} \end{pmatrix} \begin{pmatrix} u \\ w \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & B_w \end{pmatrix} \begin{pmatrix} \dot{u} \\ \dot{w} \end{pmatrix} = \begin{pmatrix} f(d) \\ f_w \end{pmatrix}. \tag{10.2}$$

The first row is elliptic in u for given (d, w); the second is a first-order equation for w. The largest eigenvalues of $B_w^{-1} S$, with S the Schur complement on w, scale like the hydraulic diffusivity times p^4/h^2. An explicit step would have to stay below about 200 s at $h = 100$ m and $p = 2$, while quasi-dynamic steps reach 3×10^6 s (inferred estimate, plan revision 2.2, §8.7). So w is advanced implicitly.

10.2 The split step (D7)

Each quasi-dynamic step of size Δt is split: implicit diffusion over $\Delta t/2$ with slip held at d^n, then today's RK45 step for (d, ψ) with w held fixed, then diffusion over $\Delta t/2$ with slip held at d^{n+1}. Each diffusion half-step is backward Euler on (10.2):

$$\begin{pmatrix} K_E^u & Z_{uw} \\ Z_{wu} & Z_{ww} + \frac{2}{\Delta t} B_w \end{pmatrix} \begin{pmatrix} u \\ w \end{pmatrix}^* = \begin{pmatrix} f(d) \\ f_w + \frac{2}{\Delta t} B_w w^n \end{pmatrix}. \tag{10.3}$$

The matrix is symmetric positive definite: $\mathbb{K}$ is positive semi-definite and definite on u (as in today's quasi-dynamic solve), and B_w is definite on w. Inside the RK45 step, each stage solves $K_E^u u = f(d) - Z_{uw} w^*$. Its matrix is today's quasi-dynamic stiffness built with λ_u (Proposition 6.2), factorized once, and the fault sees (7.4) with $\hat{p}$ from (7.2) at (u, w^*). In code this is hook H5 in `SEASQuasiDynamicOperator::Mult` (`solver/seas_operator.hpp:318-335`), and the half-steps wrap `ode_solver.Step` in the native RK45 loop (`drivers/seas_driver.cpp:580-590`, hook H6). Matrix (10.3) depends on Δt; D8 factorizes it with MUMPS and refactorizes when Δt drifts, with the stale factorization as a PCG preconditioner in between. The source loads are evaluated at the end time of each half-step and each stage.

10.3 The switch between quasi-dynamic and dynamic (P8)

Both solvers use the same space, the same K_E^u (`ElasticityDomainOperator`, `drivers/seas_driver.cpp:329`) and the same Z blocks, so the state (u, w, d, ψ) crosses the switch by direct copy. Quasi-dynamic to dynamic: $x = [u; w]$ from the last solve, the solid velocity as the elastic switch sets it today, $\dot{w}$ = the Darcy rate of the last step, and $a^0 = 0$. Dynamic to quasi-dynamic: keep (w, d, ψ); the next quasi-dynamic solve recomputes u.

11. Where each equation comes from

"Ours" means derived in this document, with the proof cited; nothing marked ours is taken from a paper. "Code" means today's code at `5b2c77d`, reused unchanged. The first table covers the bulk equations and their SIPG form, the second the fault, the matrices and the time step.

Eq.	Content	Source	Ours?
(4.1)–(4.2)	Total momentum; relative fluid motion with Darcy drag	Biot 1956, 1962; written as Antonietti, Botti, Mazzieri 2022, review eq. (31)	No
(4.3)	Mass balance with the injection rate Q; Biot's law; the integrated form with V	Biot 1962 for the mass balance and law; Q as the general source by Chunhui Zhao's decision (F1)	The integrated form and the split into V and p_{init} (§4.2)
(4.4)	Derived constants	SeisSol `Helper.h`, lines 41–65 (master, read 20 Oct 2026); Gassmann 1951 for K_u	No
(4.5)	Energy	Standard for Biot's equations	Proof here (Proposition 4.1)
(4.6)	High-frequency P and S speeds	Biot 1956 (the high-frequency limit of the dispersion relation)	Derivation here
(4.7)	Three-field form	The note of 20 Oct 2026 (Sivaselvan et al. 2016; Mella et al. 2023, as cited there; not read here)	Equivalence proof (Proposition 4.2)
(5.1), (6.1)	Element weak form; exact face terms	Integration by parts and Lemma 3.1	Yes
(6.2)–(6.3)	SIPG form	The pattern (full jump of u, normal jump of w only) from Antonietti et al. 2022, SISC and review eqs. 32–36	The total-stress split (F2; Propositions 6.1, 6.2)
(6.4)	Solid penalty	Code: `ComputePenalty`, with λ_u	No
(6.5)	Fluid penalty	By analogy with (6.4); Antonietti et al. use a user constant (review eq. 36)	Yes (inferred, checked by tests U2-5 and U2-18)
(6.6)	Source loads	From (5.1)	Yes
(7.1)	Slip load	Code: `AssembleSlipFaceRHS`, formulation S	No
(7.2)	Fault pressure $\hat{p}$	The SIPG flux of (6.2); SeisSol instead solves a Riemann problem	Yes
(7.3)–(7.4)	Total traction; effective normal stress	Terzaghi 1943 (coefficient 1); SeisSol `FrictionSolverCommon.h`, `LinearSlipWeakening.h`; Wolf et al. 2025	The split of T into today's kernel plus the coupling term
Start of §7.4	`tpv5-poro` starting pressure	SeisSol `docs_initial-condition.rst:146-150`, `parameters.par`	No
(8.1)–(8.4)	Matrices; local mass and drag; relaxation time	From (6.7)	Yes

(9.1)–(9.2)	Newmark with $\beta = 0$	Newmark 1959; code: `CentralDifferenceSolver`	The velocity-coefficient matrix Γ
(9.3)–(9.4)	Fitted coefficient γ_w	The exponential-fitting idea	Yes (Proposition 9.1)
(9.5)	Γ_{mom}		Yes (F3; Propositions 9.2, 9.3)
(9.6)	Time-step bound	Code: `ComputeMaxDt` with c_{fast}	No
(10.1)–(10.3)	Quasi-static Biot; split step	The standard quasi-static limit of Biot's equations	The split step (D7)

References. M. A. Biot (1956), Theory of propagation of elastic waves in a fluid-saturated porous solid. I. Low-frequency range, J. Acoust. Soc. Am. 28(2) 168–178. M. A. Biot (1962), Mechanics of deformation and acoustic propagation in porous media, J. Appl. Phys. 33(4) 1482–1498. F. Gassmann (1951), Über die Elastizität poröser Medien, Vierteljahrsschr. Naturforsch. Ges. Zürich 96, 1–23. K. Terzaghi (1943), Theoretical Soil Mechanics, Wiley. N. M. Newmark (1959), A method of computation for structural dynamics, J. Eng. Mech. Div. ASCE 85(EM3) 67–94. P. F. Antonietti, M. Botti, I. Mazzieri, S. Nati Poltri (2022), SIAM J. Sci. Comput. 44(1) B1–B28. P. F. Antonietti, M. Botti, I. Mazzieri (2022), Vietnam J. Math. 50(4) 997–1028 (arXiv:2111.02311). S. Wolf, M. Galis, C. Uphoff, A.-A. Gabriel, P. Moczo, D. Gregor, M. Bader (2022), J. Comput. Phys. 455, 110886. S. Wolf, A.-A. Gabriel, M. Galis, P. Moczo, D. Gregor, M. Bader (2025), Geophys. J. Int. 242(2) ggaf184. DOIs and summaries are in `/mnt/project-files/literature/poroelastic/LITERATURE_NOTES.md`.

12. Where it lives

- This document: `/mnt/project-files/poroelastic/derivation-2026-10-20/poro-derivation-rev1.2.pdf`, with its source HTML beside it. Revisions 1 and 1.1 are kept in the same folder.
- The plan it accompanies: `/mnt/project-files/poroelastic/plan-2026-10-20/poro-plan-rev3.pdf`. It takes its equations from this document (F1 as decided, F2 and F3 as recommended) and adds the implementation steps, the unit tests and the verification tests. Its reference values come from `poro_test_refs.py` in the same folder.
- Branch `nitsche-dynamic-rupture-poroelastic` at `5b2c77d`. Code cited: `integrator/dg_elasticity_ip_combined_integrator.hpp`, `integrator/dg_absorbing_bc_integrator.hpp`, `domain/elasticity_operator.hpp`, `domain/nitsche_wave_operator.hpp`, `domain/nitsche_mass.hpp`, `domain/nitsche_material_sample.hpp`, `solver/nitsche_cfl.hpp`, `solver/seas_operator.hpp`, `drivers/seas_driver.cpp`, `drivers/nitsche_dyn_driver.cpp`, `tpv205/configs/tpv205_nitsche.toml`, all under `miniapps/seas/`; for example dg_elasticity_ip_combined_integrator.hpp.
- SeisSol: `Helper.h` (master, read 20 Oct 2026) for the constants; `FrictionSolverCommon.h`, `LinearSlipWeakening.h`, `docs_initial-condition.rst`, and the `tpv5-poro` inputs `parameters.par`, `material.yaml`, `fault.yaml`, all in `/mnt/project-files/literature/poroelastic/tpv5-poro/`.